

Event Volatility

Pricing the jump from the term structure, and trading it with a bounded tail

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Study notes

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Abstract

An option whose life spans an information event (an earnings release, a central-bank decision, a binary ruling) carries an implied variance that splits, under a standard jump-diffusion model, into a diffusion part and an event part. These notes build everything on that one fact.

Part I prices the event. It derives the decomposition, extracts the implied event volatility from the term structure, explains the mechanical rise and collapse of implied volatility around the event, and shows why that pattern is not a tradable signal under fair pricing. What can be traded is the *level* of event volatility against its realized history.

Part II trades it. Selling a short-dated straddle into the event earns that premium with an unbounded left tail. Selling the same short-dated straddle against a longer-dated one at the same strike, with both alive through the event, turns the same view into a position whose loss through the event is bounded by the debit paid. In a worked example, repriced exactly and checked by simulation, the bound costs most of the edge: the calendar keeps about a tenth of it, and the tenor of the long leg is the dial that trades edge against tail. Because that edge is small, the calendar also needs cheap execution: in the example it breaks even at a cost of about one percent per side, against nineteen for the naked straddle.

The single observation underlying all of this is that *variance is additive*, while volatility is not.

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Part I

Pricing the Volatility of an Event

1. Motivation: the puzzle of the IV path

A stock reports earnings on Thursday after the close. A trader records the implied volatility of the ATM option expiring on Friday, each morning of the week:

Day	Implied vol (ATM, Friday expiry)
Monday	10.0%
Tuesday	10.5%
Wednesday	11.2%
Thursday (event after the close)	12.5%
Friday	8.0%

Implied volatility climbs into the event and collapses afterwards. The temptation is overwhelming: buy volatility on Monday and sell it on Thursday after the run-up, or sell the straddle before the release and harvest the collapse. The puzzle is that, under fair pricing, none of these trades makes money in expectation.

These numbers are not an empirical anecdote. They are exactly what the decomposition of §2 produces with an 8% diffusion volatility and a 0.85% event, and §4 reproduces them line by line. That is the point: the pattern needs no explanation beyond arithmetic. The goal of Part I is to make that arithmetic explicit, and to say what is left to trade once it is understood.

The retail framing. “Vol goes up before earnings, sell vol after earnings.” This is not wrong as a description; it is wrong as a trading rule. The path is a consequence of an identity, not a signal of mispricing.

The professional framing. “The implied event volatility is X ; realized event moves on this name have run at Y ; is X rich or cheap relative to Y ?” This is the central event-volatility question; others, such as the asymmetry of the event or how it transmits across markets, build on it.

2. The variance decomposition

2.1 A continuous process punctuated by a jump

Let the underlying S_t follow a diffusion with volatility σ_{diff} , and at the event time τ_e take a discrete log-jump J with $\mathbb{E}[J] \approx 0$ and $\text{Var}(J) = \sigma_{\text{event}}^2$ under the pricing measure:

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_{\text{diff}} dW_t + (e^J - 1) dN_t, \quad (1)$$

where N_t is a counting process that fires once, at τ_e .

2.2 Total variance over the option's life

For an option with time to expiry T (in years), the log-return is the sum of a diffusion and, if the event falls within the option's life, a jump:

$$\ln(S_T/S_0) = \underbrace{cT + \sigma_{\text{diff}} W_T}_{\text{diffusion}} + \underbrace{J \mathbf{1}[\tau_e \leq T]}_{\text{event}}, \quad (2)$$

where c is a constant drift and the indicator equals one when the scheduled event falls before expiry. The two pieces are independent, so the variance of the log-return adds across them:

$$\text{Var}[\ln(S_T/S_0)] = \sigma_{\text{diff}}^2 T + \sigma_{\text{event}}^2 \mathbf{1}[\tau_e \leq T]. \quad (3)$$

The Black–Scholes implied volatility σ_{IV} is defined by matching the option's price, not its variance; to leading order, it is the constant volatility that produces the same total variance:

$$\sigma_{\text{IV}}^2 T = \sigma_{\text{diff}}^2 T + \sigma_{\text{event}}^2. \quad (4)$$

The variance decomposition. For an option with T years to expiry whose life contains an event,

$$\sigma_{\text{IV}}^2 T = \sigma_{\text{diff}}^2 T + \sigma_{\text{event}}^2$$

Total implied variance is diffusion variance plus event variance. Variance adds across independent sources; volatility does not.

Remark. The two terms are different kinds of variance. The diffusion term accumulates: σ_{diff} is an annualized rate, and the variance it delivers grows with the time it has to act. The event term is discrete: σ_{event} is the standard deviation of a single move, not a rate, so σ_{event}^2 is a lump of variance delivered at once, the same for any option whose life contains the event. The decomposition adds two totals measured over the same horizon: what the diffusion accumulates over the option's life, and what the event delivers in one night. They add because the two pieces are uncorrelated, which is all the addition needs: for centred variables, variance is a squared norm and covariance an inner product, so the additivity of variance is Pythagoras' theorem. It is also why volatilities, the norms themselves, never add. Expressed as an annualized implied volatility, the lump is spread over the option's life,

$$\sigma_{\text{IV}}^2 = \sigma_{\text{diff}}^2 + \frac{\sigma_{\text{event}}^2}{T},$$

and that division by T is what drives the path of implied volatility in §4.

The additivity of variance is exact. Its translation into a Black–Scholes implied volatility is exact only when the log-jump is normal: the terminal log-return is then itself normal with variance $\sigma_{\text{diff}}^2 T + \sigma_{\text{event}}^2$, so a Black–Scholes price at σ_{IV} is the jump-diffusion price, at every strike. For any other jump law the log-return is not normal, the smile is no longer flat, and the at-the-money volatility only approximates the variance-matched one: closely when the event is small against the diffusion, poorly when it dominates. For a binary event, with two possible outcomes, the at-the-money volatility can exceed the variance-matched one by up to 25%. The normal jump is also what lets the fair-pricing check of §11 price with Black–Scholes at a single volatility and still come out at zero.

2.3 Solving for the event volatility

$$\sigma_{\text{event}} = \sqrt{(\sigma_{\text{IV}}^2 - \sigma_{\text{diff}}^2) T}. \quad (5)$$

The event volatility is the square root of the variance *beyond* what diffusion alone would produce. The task is to extract σ_{IV} and σ_{diff} from the market and infer σ_{event} .

3. Extracting the event volatility: the term-structure cliff

3.1 Bracketing expiries

The cleanest extraction compares two options at the same strike whose lives differ only in whether they contain the event:

- **Option A** expires *before* the event. Its implied volatility is pure diffusion: $\sigma_{\text{IV}}^A = \sigma_{\text{diff}}$.
- **Option B** expires *after* the event: $(\sigma_{\text{IV}}^B)^2 T_B = \sigma_{\text{diff}}^2 T_B + \sigma_{\text{event}}^2$.

Substituting $\sigma_{\text{diff}} = \sigma_{\text{IV}}^A$ into the second equation,

$$\sigma_{\text{event}}^2 = T_B \left[(\sigma_{\text{IV}}^B)^2 - (\sigma_{\text{IV}}^A)^2 \right]. \quad (6)$$

Do not difference the quoted total variances. It is tempting to write $\sigma_{\text{event}}^2 = (\sigma_{\text{IV}}^B)^2 T_B - (\sigma_{\text{IV}}^A)^2 T_A$. That quantity is not the event variance: it also contains the diffusion accrued between the two expiries, $\sigma_{\text{diff}}^2 (T_B - T_A)$, and it overstates the event. With $\sigma_{\text{diff}} = 20\%$, a true $\sigma_{\text{event}} = 6.00\%$, and expiries two and seven trading days out, the naive formula reads 6.63%: ten percent too high in volatility, twenty-two percent in variance.

Intuition. Plot implied total variance against tenor. Before the event it is a straight line of slope σ_{diff}^2 ; at the event date it jumps up by σ_{event}^2 ; after it, it continues with the same slope. The event variance is the *height of that jump*, the vertical gap between the two line segments at a common tenor, not the difference between two points at different tenors. This is the term-structure cliff (Figure 1).

3.2 When both expiries contain the event

Many names have no listed expiry between today and the event: with monthly options only, every available expiry contains it. Two such expiries still identify both unknowns. With

$$\sigma_1^2 T_1 = \sigma_{\text{diff}}^2 T_1 + \sigma_{\text{event}}^2, \quad \sigma_2^2 T_2 = \sigma_{\text{diff}}^2 T_2 + \sigma_{\text{event}}^2, \quad (7)$$

differencing cancels the event, because both lives contain the same jump:

$$\sigma_{\text{diff}}^2 = \frac{\sigma_2^2 T_2 - \sigma_1^2 T_1}{T_2 - T_1}, \quad \sigma_{\text{event}}^2 = \sigma_1^2 T_1 - \sigma_{\text{diff}}^2 T_1. \quad (8)$$

The two legs jointly determine the diffusion and the event, with no pre-event expiry required. They also provide a consistency check: if a diffusion volatility estimated elsewhere disagrees with (8), the two legs return different event volatilities, and the disagreement is the signal that the estimate is wrong (§11 shows a concrete case). The method assumes the diffusion volatility is flat between the two expiries; if it is not, the slope is absorbed into the event estimate, which is one more reason to prefer close expiries.

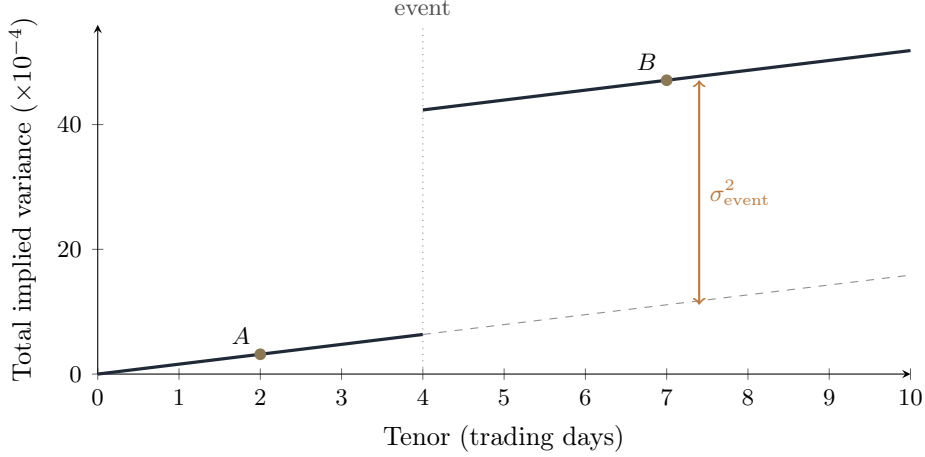


Figure 1: The term-structure cliff, with $\sigma_{\text{diff}} = 20\%$ and $\sigma_{\text{event}} = 6\%$ and the event after day four. Total variance grows along a line of slope σ_{diff}^2 and jumps by σ_{event}^2 at the event. The event variance is the gap between the two segments at a common tenor (arrow), not the difference between the quotes at A and B, which also contains the diffusion accrued between them.

3.3 Practical considerations

1. **Strike.** Use the same strike for both options, typically ATM-forward, where the implied volatility is least contaminated by skew. The decomposition holds at every strike, but ATM quotes are the most reliable.
2. **Proximity.** The expiries should bracket the event closely, so that the diffusion volatility has little room to drift between them.
3. **Liquidity.** For large capitalizations, weekly options are usually liquid around earnings. For less liquid names the cliff is noisier, and wide quotes should be filtered before extraction.
4. **Multiple events.** If two events fall between the expiries, the extracted variance is their sum. Separating them requires a third expiry that brackets only one.

4. The mechanical IV path

4.1 Why implied volatility rises into the event

Hold the option fixed and let time pass. The event variance σ_{event}^2 is a fixed quantity, but expressed as a volatility it is divided by the remaining time $\tau(t)$:

$$\sigma_{\text{IV}}^2(t) = \sigma_{\text{diff}}^2 + \frac{\sigma_{\text{event}}^2 \mathbb{1}[\tau_e > t]}{\tau(t)}. \quad (9)$$

While the event is ahead, the second term grows as its denominator shrinks. *The rise requires no new information.* The same event variance is being concentrated into fewer remaining days. Real implied volatility also moves as the market revises its view of the event, but the mechanical rise would be there without it.

4.2 Why it collapses afterwards

Once the event has passed, the indicator switches off. Whatever the jump was, it is now in the spot price, and the option's remaining life is pure diffusion:

$$\sigma_{IV}^2(t) = \sigma_{\text{diff}}^2 \quad \text{for } t > \tau_e. \quad (10)$$

The collapse is the disappearance of the event term.

4.3 Worked example

With $\sigma_{\text{diff}} = 8\%$ and $\sigma_{\text{event}} = 0.85\%$, a Friday-expiry option observed each morning gives:

Day	τ (years)	Event ahead?	σ_{IV}
Monday	5/252	yes	$\sqrt{0.08^2 + 0.0085^2/(5/252)} = 10.0\%$
Tuesday	4/252	yes	$\sqrt{0.08^2 + 0.0085^2/(4/252)} = 10.5\%$
Wednesday	3/252	yes	$\sqrt{0.08^2 + 0.0085^2/(3/252)} = 11.2\%$
Thursday	2/252	yes	$\sqrt{0.08^2 + 0.0085^2/(2/252)} = 12.5\%$
Friday	1/252	no	$\sqrt{0.08^2} = 8.0\%$

This is the table of §1, reproduced from two parameters, and Figure 2 shows the same path continuously. Quotes taken later on Thursday sit higher still, because τ keeps shrinking during the session: by Thursday's close, with one day left, the same formula gives 15.7%. A table of morning quotes understates the peak, not the mechanism.

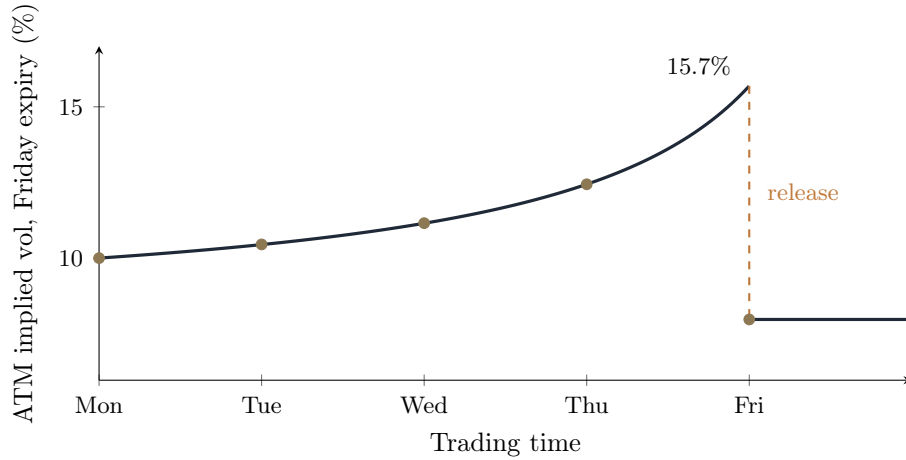


Figure 2: The implied volatility of the Friday-expiry option through the week, in trading time, with $\sigma_{\text{diff}} = 8\%$ and $\sigma_{\text{event}} = 0.85\%$. The dots are the morning quotes of §1. Volatility rises along $\sqrt{\sigma_{\text{diff}}^2 + \sigma_{\text{event}}^2/\tau}$ as the remaining time τ shrinks, peaks at 15.7% just before the release after Thursday's close, and falls to the diffusion level of 8% once the event has passed.

5. Why the IV path is not a tradable signal

5.1 The P&L of a delta-hedged option

From the variance–P&L decomposition of the companion notes on hedged straddles, a continuously delta-hedged long option earns

$$\Pi \approx \frac{1}{2} \int_0^T \Gamma_t S_t^2 [(\sigma_{\text{real},t})^2 - \sigma_{\text{IV}}^2] dt : \quad (11)$$

realized variance against implied variance, weighted by dollar gamma.

5.2 Zero expected P&L under fair pricing

Call the event fairly priced when its implied variance equals the expected realized variance, so that no event risk premium is charged. Then

$$\mathbb{E} \left[\int_0^T (\sigma_{\text{real},t})^2 dt \right] = \sigma_{\text{IV}}^2 T = \sigma_{\text{diff}}^2 T + \sigma_{\text{event}}^2, \quad (12)$$

with the diffusion delivered continuously and the event as a single squared jump. Both are paid for in the premium, and the expected P&L of the hedged option is zero. Under the pricing measure this would hold by construction; the substantive claim is that it holds under the real distribution of outcomes when no premium is charged, and (11) shows how the pieces offset. Part II checks this numerically: with a fairly priced event, both a short straddle and a calendar have expected P&L of zero to Monte-Carlo precision (§11).

The IV path is not a free profit. The rising implied volatility before the event is paid for in the premium the seller collects; the collapse after it is paid for by the gamma realized at the event itself. The two cancel in expectation. There is no free lunch in a mechanical pattern that the market also observes and prices.

The cancellation is exact in expectation, not path by path: when the jump lands, the dollar gamma it meets depends on where spot happens to be, which is the path-dependence of the straddle studied in the companion notes.

5.3 The shape of the P&L distribution

Zero expected P&L does not mean a symmetric one. Selling event volatility wins most of the time, when the move is smaller than priced, and loses rarely but heavily, when it is much larger. Part II quantifies this: with the event priced exactly right, a short straddle through it still wins 59% of the time, at an expected P&L of zero. A backtest over a quiet sample will make it look attractive. Selling event volatility without a view on its mispricing is selling lottery tickets one has not priced.

5.4 Where the confusion comes from

The temptation rests on a real observation: vega changes mechanically as the remaining time to the event shrinks. Marked to the implied path, a vega position shows real P&L as volatility rises into the event and falls after it. The catch is that the gamma runs the other way: short volatility into the event is short gamma into the highest-variance moment of the option's life. A full accounting of vega, gamma and theta shows the pieces offset in expectation.

Intuition. Vega and gamma are not independent. They are tied by the identity $\nu = S^2 \sigma T \Gamma$. An option rich in vega before the event is also rich in gamma, and a short position in it carries the gamma risk that pays, in expectation, for the vega gain.

6. What can be traded

If the path is not a signal, the level can be. Extract the implied event volatility from the cliff, and compare it with your estimate of the event risk. That estimate may incorporate historical event moves, fundamental information, comparable events, or other evidence relevant to the specific event.

- If the implied event volatility is materially *above* realized, the market is overpaying for the event: sell event volatility.
- If it is materially *below*, the market is underpricing the event: buy it.

The P&L of such a position is, to leading order, the realized squared jump against the implied event variance.

The obvious vehicle for the short side is a straddle sold into the event. Its distribution is the problem of §5: a high win rate over an unbounded left tail. Part II develops the structure that keeps the same view with a floor under it.

One further layer is worth flagging. Events shape the smile as well as its level. A skewed event tilts it. A binary event, with two distinct outcomes, makes the density bimodal: on its own it bends the smile into a concave hump at the money, and on a base smile with convex wings it produces the W-shaped curves sometimes seen before such events (Glasserman and Pirjol, 2023). Concave curves are frequent before earnings announcements, and they signal both larger moves and more expensive protection (Alexiou et al., 2025). SVI, being convex, cannot fit these shapes; the future notes on binary events show how to fit and trade them.

7. Calibration in practice

7.1 Diffusion volatility

Estimate σ_{diff} from options that do not span the event, or from realized volatility over quiet periods:

- the ATM implied volatility of an expiry before the event;
- realized volatility over the prior N trading days, excluding earlier events;
- range-based estimators (Garman–Klass, Parkinson) for higher resolution;
- or, when no pre-event expiry exists, the two-leg solution (8).

7.2 Implied event volatility

With a pre-event expiry, use the cliff (6); without one, use the two-leg solution (8). Where several expiries bracket the event, compute both and compare: agreement is the check that the diffusion assumption holds. Discard outliers driven by wide or stale quotes.

7.3 Realized event volatility

The implied event variance is the market’s price of $\mathbb{E}[J^2]$, the average *squared* move. Its realized counterpart is therefore the average of past squared moves, and its square root, the root-mean-square move, is the realized event volatility.

The simplest construction records, for each past event i , the close-to-close move across the release,

$$J_i = \ln(S_{\text{post-event close}}/S_{\text{pre-event close}}), \quad (13)$$

and takes

$$\hat{\sigma}_{\text{event}} = \sqrt{\frac{1}{N} \sum_{i=1}^N J_i^2}. \quad (14)$$

It is one construction among several. The close-to-open move isolates the overnight gap in which a release outside trading hours is first priced, at the cost of missing any continuation in the following session; subtracting the index's move over the same window removes the market's own contribution. Each choice trades noise against completeness, and what matters most is to make it once and hold it fixed across events.

Two tempting substitutes, and why they fail.

- **The average absolute move** is a different quantity, and a smaller one: for a normal jump, $\mathbb{E}|J| \approx 0.80\sigma$. Compared with the implied volatility, it makes a normal event look about a quarter more expensive than it is, and a fat-tailed one more so. On the eight moves of §11, the average move is 3.65%, against an RMS of 3.89%.
- **The standard deviation of the move sizes** measures how much the moves differ from one another, not how large they are. A stock that moved exactly 5% at every release would have a dispersion of zero and an event volatility of 5%. On the same eight moves it is 1.44%: not an event volatility at all.

Two corrections complete the estimate. A close-to-close move contains one session of ordinary diffusion as well as the event, and its variance can be removed,

$$\hat{\sigma}_{\text{event}}^2 = \frac{1}{N} \sum_{i=1}^N J_i^2 - \sigma_{\text{diff}}^2 \Delta t, \quad (15)$$

with Δt one trading day. And the sample is small. Each squared move is a very noisy number: for normal moves, its standard deviation is $\sqrt{2}$ times its average. Averaging N of them divides that noise by $\sqrt{N}$, so the estimated variance carries a relative error of about $\sqrt{2/N}$, and the volatility, its square root, about half that. With eight events that is 50% in variance and 25% in volatility: a true event volatility of 4% is measured between 2.9% and 4.9% two times out of three, and anywhere from 2.3% to 5.6% nine times out of ten. Normal moves are the best case; fat tails make the error larger. Only large gaps against the implied value are informative.

7.4 The signal

$$\text{Signal} = \frac{\sigma_{\text{event}}^{\text{implied}}}{\hat{\sigma}_{\text{event}}^{\text{realized}}}. \quad (16)$$

Values well above 1 suggest the event is overpriced; well below, underpriced. Given the estimation noise just described, a ratio near 1 is inside the noise, and the signal is most useful relative to its own history on the name rather than in absolute terms.

The history may describe a different company. The comparison treats past events as draws from one distribution, and they rarely are. What a release can reveal depends on the company's situation at the time: a pending acquisition, a change of strategy or of guidance policy, new leverage, or a different market regime can make this event larger or smaller than any in the sample. The market prices this event, not the average of past ones, so a gap between implied and realized history is a question rather than a verdict: either the event is mispriced, or the market knows it is different. Weighting recent events more heavily, excluding those from a business that no longer exists, and comparing with the events of similar companies all help; none of them removes the need for judgement.

Part II

Trading the Event: the Post-Event Calendar

Assumptions for Part II. European options, zero interest rates and no dividends; options marked at Black–Scholes prices; a flat diffusion volatility between the expiries involved; a normal log-jump at the event, used for the simulations; after the event, both legs trading at the same implied volatility, the diffusion level; and at-the-money straddles, one short against one long, at the same strike. Words such as “never” and “exactly” in what follows hold within these assumptions, and §12 discusses what happens when they fail.

8. Why the naked short straddle is dangerous

The standard short event trade sells a short-dated straddle into the event and buys it back afterwards, collecting the collapse of implied volatility. It works in expectation only if the implied event volatility exceeds the realized one, and its distribution is severely asymmetric: many small wins when the move is smaller than priced, and rare losses that grow without limit when it is larger.

The example of §11 puts numbers on it. There the market prices an event volatility of 6.17%. If that is exactly right, so that the moves that follow have the same volatility, the naked seller wins 59% of the time at an expected P&L of zero, and the worst outcomes run to several times the premium collected. If the event is overpriced, with realized moves of 3.89% against the 6.17% implied, a realized variance of about 40% of the implied, the win rate rises to 80% and the expected P&L turns positive, but the 1% loss is still about the size of the whole premium.

The arithmetic of negative skew. A trade that wins 75% of the time with average win W , and loses 25% of the time with average loss L , has expected P&L $0.75W - 0.25L$. If $L = 5W$ the expectation is $-0.5W$, negative, despite the high win rate. A long run of wins says little about expected value until the sample is large enough to contain the tail.

The structural problem is that the naked seller absorbs the event jump through gamma with nothing absorbing it in the opposite direction. Once the position is open, that risk cannot be removed without buying back optionality.

9. The post-event calendar

9.1 The trade

The calendar has two legs, both opened before the event and both at the same strike:

- **Short** a straddle expiring shortly after the event;
- **Long** a straddle expiring further out.

Both lives contain the event, and both options are alive through it. The position is held from

before the release until shortly after; Figure 3 lays out the dates.

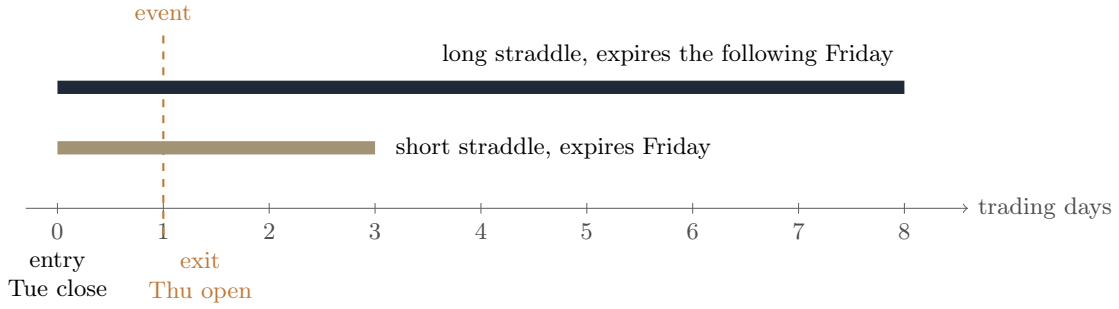


Figure 3: The calendar of the worked example on a trading-day axis. Both straddles are traded at Tuesday's close and both remain alive through the release after Wednesday's close; the position is closed at Thursday's open. The short straddle expires three trading days after entry, the long one eight.

9.2 It is always a debit

Both legs contain the event, and the longer one carries more diffusion on top of it, so its total variance is larger and it costs more. Absent a calendar arbitrage (total implied variance falling with tenor), a same-strike calendar is always paid for. Write the debit as

$$D = C_{\text{long}}^{\text{entry}} - C_{\text{short}}^{\text{entry}} > 0. \quad (17)$$

9.3 What the calendar is short

Both legs contain the *same* event variance, so it is tempting to think the event cancels out of the calendar entirely. It does not quite. At zero rates, a straddle's value depends on its total variance $w = \sigma^2 T$ alone, and roughly as $0.8 S \sqrt{w}$. The square root is concave: the first units of variance add more value than the later ones, as the first spoonful of sugar sweetens a cup more than the tenth. Its slope is

$$\frac{\partial C}{\partial w} = \frac{1}{2} S^2 \Gamma, \quad (18)$$

where $S^2 \Gamma$ is the dollar gamma of the straddle, and it is larger when w is small. The same event variance is therefore worth more to the short-dated leg, which has little variance of its own, than to the long-dated one.

Both legs contain the *same* event variance, so it is tempting to think the event cancels out of the calendar entirely. It does not quite, because an option's value responds to variance with diminishing returns. At zero rates a straddle is worth roughly $0.8 S \sqrt{w}$, the square root of its total variance $w = \sigma^2 T$: the first units of variance add more value than the later ones, as the first spoonful of sugar sweetens a cup more than the tenth. The short-dated leg has little variance of its own, so the event adds more to it than to the long-dated leg.

The example of §11 shows how much more. If the market raised its estimate of the event variance by 10%, the short straddle would gain 0.45 and the long straddle 0.42. The naked seller, short the first, loses the 0.45. The calendar, short the first and long the second, loses only the difference, 0.04: it keeps about 8% of the naked straddle's exposure to the event.

Where the numbers come from.

The slope. Differentiating $C \approx 0.8 S \sqrt{w}$, and using $\frac{d}{dw} \sqrt{w} = \frac{1}{2\sqrt{w}}$,

$$\frac{\partial C}{\partial w} \approx \frac{0.4 S}{\sqrt{w}},$$

the dollars of value that a unit of total variance adds to a straddle.

The slopes in the example. Each leg's total variance w is its diffusion plus the event variance 0.00381, and $S = 200$. The table gives the slope of each leg (the exact derivative of the straddle's price, within half a percent of $0.4S/\sqrt{w}$), and what a 10% rise in the event variance, $\Delta w = 0.000381$, does to the leg:

	w	$\sqrt{w}$	Slope $\partial C/\partial w$	Slope $\times \Delta w$	Exact repricing
Short leg	0.00429	0.0655	1218	0.46	0.45
Long leg	0.00508	0.0713	1119	0.43	0.42

"To first order" means using only the slope at the starting point: the change in value is the slope times Δw . Exact repricing recomputes the straddle at the new variance. The first-order figure is slightly larger because the curve bends: the slope falls as variance is added, so the starting slope overstates the average slope over the step.

The link with gamma. In one line, with $w = \sigma^2(T - t)$ and the Black–Scholes equation at zero rates,

$$-\sigma^2 \frac{\partial C}{\partial w} = \frac{\partial C}{\partial t} = -\frac{1}{2} \sigma^2 S^2 \Gamma \implies \frac{\partial C}{\partial w} = \frac{1}{2} S^2 \Gamma.$$

The first equality is the chain rule, since C depends on time only through w ; the second is the Black–Scholes equation. This is the same $\frac{1}{2} S^2 \Gamma$ that multiplies realized variance in the P&L of (11), and the form used in §10.

Intuition. The naked seller is short the event's variance at full exposure. The calendar is short only the difference in how two options price the same event: a far smaller exposure, with a hard floor beneath it.

10. The P&L through the event

10.1 An exact decomposition

After the release, the event has left the remaining life of both options, and both implied volatilities collapse to the diffusion level σ_{diff} . Holding from entry to just after the event, the P&L is

$$\Pi = (C_{\text{short}}^{\text{entry}} - V_{\text{short}}^{\text{post}}) + (V_{\text{long}}^{\text{post}} - C_{\text{long}}^{\text{entry}}) = -D + (V_{\text{long}}^{\text{post}} - V_{\text{short}}^{\text{post}}). \quad (19)$$

The first term is fixed at entry. Everything the event does to the position lives in the second.

Notation	Meaning
$C_{\text{short}}^{\text{entry}}$	Short-leg value at entry.
$C_{\text{long}}^{\text{entry}}$	Long-leg value at entry.
$V_{\text{short}}^{\text{post}}$	Short-leg value after the event.
$V_{\text{long}}^{\text{post}}$	Long-leg value after the event.
D	Value of the premium paid for $C_{\text{short}}^{\text{entry}} - C_{\text{long}}^{\text{entry}}$.

Result 10.1 (The loss is bounded by the debit). *If both legs are struck at the same K and trade at the same post-event volatility, then, under the remaining assumptions of Part II, $\Pi \geq -D$ for every realized move, and $\Pi \rightarrow -D$ as the move grows without limit. The bound does not depend on the distribution of the jump.*

Proof. After the event, at the same spot, strike and volatility, the two straddles differ only in their remaining total variance, and a straddle's value increases with total variance (its vega is positive). Hence $V_{\text{long}}^{\text{post}} \geq V_{\text{short}}^{\text{post}}$ for every spot, and $\Pi \geq -D$. For a very large move, both values are dominated by the same intrinsic value $|S - K|$, their time values vanish, and the difference tends to zero, so $\Pi \rightarrow -D$. $\square$

The gain is bounded too. The time-value difference $V_{\text{long}}^{\text{post}} - V_{\text{short}}^{\text{post}}$ is largest when spot finishes near the strike, so the calendar's P&L lives in a fixed interval: at worst the debit, at best the debit plus the at-the-money time-value difference. The naked straddle has no such floor; its loss grows linearly with the move.

10.2 Expected P&L

To leading order, the expected P&L is the event exposure of §9.3 times the mispricing of event variance:

$$\mathbb{E}[\Pi] \approx \frac{1}{2} (\Gamma_{\text{short}} - \Gamma_{\text{long}}) S^2 \left[(\sigma_{\text{event}}^{\text{implied}})^2 - (\sigma_{\text{event}}^{\text{realized}})^2 \right]. \quad (20)$$

It is positive when the event is overpriced, and it scales with the gamma *difference*, which is small precisely because both legs contain the event. The formula gives the sign and the mechanism, not the size. In the example of §11 it gives 0.23 for the calendar against an exact 0.38, and 2.80 for the naked straddle against 3.33: the realized event variance there is far from the implied one, and option values are concave in variance, so the first-order term understates the gain.

11. Worked example

Every number below is computed by exact Black–Scholes repricing, with the expectations estimated over two million simulated events.

11.1 Setup

A large-cap stock trades at $S = 200$ and reports earnings on Wednesday after the close. The trade is entered at Tuesday's close:

- **Short leg:** the straddle expiring Friday of the same week, $T_{\text{short}} = 3/252$, quoted at $\sigma_{\text{short}} = 60\%$;
- **Long leg:** the straddle expiring Friday of the following week, $T_{\text{long}} = 8/252$, quoted at $\sigma_{\text{long}} = 40\%$.

Both are struck at $K = 200$. The position is evaluated at Thursday's open, after the release, with one trading day elapsed.

11.2 Extracting the diffusion and the event from the two legs

Both expiries contain the event, so the two-leg solution (8) applies:

$$\sigma_{\text{diff}}^2 = \frac{\sigma_2^2 T_2 - \sigma_1^2 T_1}{T_2 - T_1} = \frac{0.40^2 \cdot \frac{8}{252} - 0.60^2 \cdot \frac{3}{252}}{\frac{5}{252}} = 0.04, \quad (21)$$

$$\sigma_{\text{event}}^2 = \sigma_1^2 T_1 - \sigma_{\text{diff}}^2 T_1 = 0.60^2 \cdot \frac{3}{252} - 0.04 \cdot \frac{3}{252} = 0.00381, \quad (22)$$

so $\sigma_{\text{diff}} = 20.0\%$ and $\sigma_{\text{event}} = 6.17\%$, and the long leg returns the same event volatility exactly.

A consistency lesson. Had the diffusion been taken from elsewhere at 25%, the short leg would imply an event volatility of 5.95% and the long leg 5.56%. The temptation is to call the gap normal noise and average it away. It is not noise: 25% is simply inconsistent with the two quotes, which jointly imply 20%. When the two legs disagree, the diffusion input is what is wrong.

11.3 Against realized history

Suppose the last eight earnings moves on the name, in absolute terms, were

$$\{2.1\%, 4.2\%, 1.8\%, 6.1\%, 3.4\%, 2.9\%, 5.0\%, 3.7\}\%.$$

Their RMS (14) is 3.89%. Against the implied 6.17% the ratio is 1.59: a wide gap, but statistically fragile on a sample of eight events. If the market's 6.17% were exactly right, the probability of drawing an eight-event RMS this low is roughly one in thirteen for normal moves, and nearly one in five for fat-tailed moves, which easily hide their true variance outside small samples. The signal points to an overpriced event and supports a short position, but this statistical uncertainty dictates conservative position sizing.

11.4 Entry

Exact Black–Scholes gives $C_{\text{short}}^{\text{entry}} = 10.45$ and $C_{\text{long}}^{\text{entry}} = 11.37$, a debit of $D = 0.93$. (The Brenner–Subrahmanyam shortcut $0.8\sigma\sqrt{T}$ gives 10.47 and 11.40, close enough for mental arithmetic.)

11.5 Outcomes by size of move

At Thursday's open both implied volatilities stand at $\sigma_{\text{diff}} = 20\%$. The table shows the post-event straddle values and the P&L of each trade, per straddle, for **upward** moves of increasing size. The moves are log-moves: +10% takes the stock from 200 to $200e^{0.10} = 221.0$, a rise of 21.0 dollars. Downward moves give almost the same numbers for the calendar. For the naked straddle they give slightly smaller losses, because the same log-move down covers fewer dollars: −10% takes the stock to 181.0, a fall of 19.0, and a short straddle loses roughly the dollar distance the stock travels from the strike. That difference of 2.0 is exactly the gap between the losses at the two ends, −8.59 at −10% against −10.59 at +10%.

Up-move	$V_{\text{short}}^{\text{post}}$	$V_{\text{long}}^{\text{post}}$	Naked P&L	Calendar P&L
0%	2.84	5.32	+7.60	+1.55
2.0%	4.51	6.31	+5.93	+0.87
3.7%	7.59	8.45	+2.86	−0.06
6.2%	12.73	12.90	−2.28	−0.75
8.0%	16.66	16.70	−6.21	−0.89
12.0%	25.50	25.50	−15.05	−0.93
20.0%	44.28	44.28	−33.84	−0.93

Result 10.1 is visible in the last column: however large the move, the calendar never loses more than the 0.93 debit, while the naked loss keeps growing. The price is equally visible in the first rows: when the move is small, the naked seller keeps up to 7.60 and the calendar at most 1.55. Near the typical historical move of 3.7% the calendar is roughly flat, and at the implied move of 6.2% both lose. Figure 4 shows the whole profile.

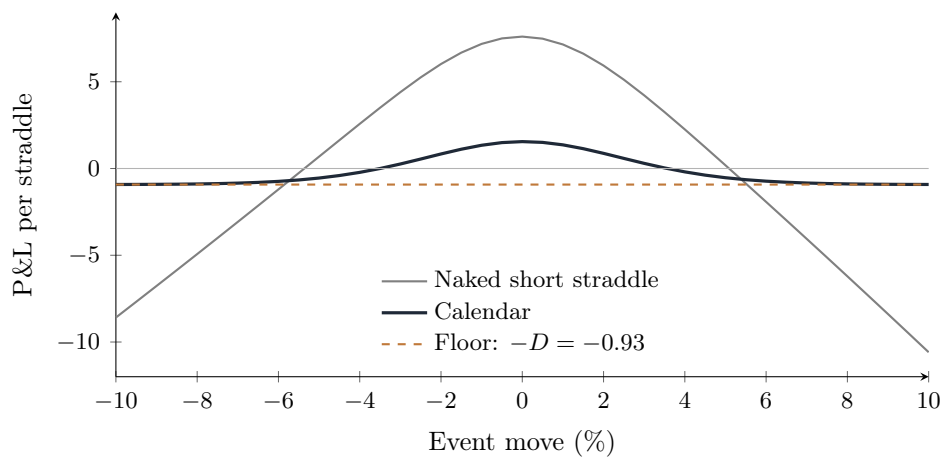


Figure 4: P&L at the post-event open against the size of the move, per straddle, for the example of §11. The naked short straddle earns most when the move is small and loses without limit as it grows. The calendar earns less, and never loses more than the debit.

11.6 Expected P&L and risk

Simulating one session of diffusion plus a normal log-jump, first with the event priced exactly (6.17%) and then with realized event volatility at its historical 3.89%:

Event	Trade	$\mathbb{E}[\Pi]$	SD	1% outcome	Worst	P(win)
Fairly priced	Naked	0.00	7.21	−22.23	−70.11	59%
	Calendar	0.00	0.89	−0.93	−0.93	43%
Overpriced	Naked	+3.33	4.40	−10.69	−32.29	80%
	Calendar	+0.38	0.86	−0.92	−0.93	61%

The tail columns rest on a normal jump. Real event moves have fatter tails, so the naked straddle's 1% and worst outcomes are understated here; the calendar's floor does not depend on the shape of the jump, so fatter tails only strengthen its case. Figure 5 shows the two distributions in the fairly priced case.

Three readings.

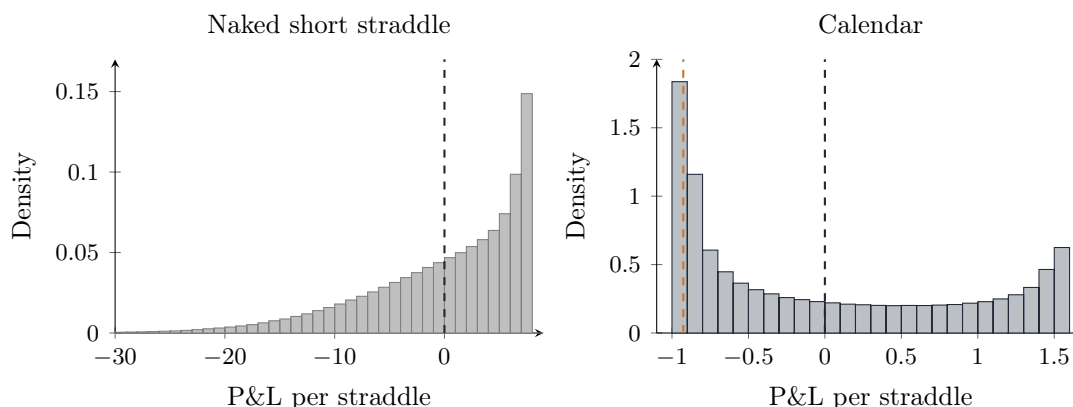


Figure 5: Distribution of the P&L through the event when the event is fairly priced, over two million simulated events. Both have an expected P&L of zero (dashed line). The naked straddle wins 59% of the time but carries a long left tail; the 0.2% of outcomes beyond -30 are not shown, and the worst reaches -70 . The calendar wins only 43% of the time, but its outcomes are confined between the floor of -0.93 (orange), where more than a fifth of them accumulate, and $+1.55$. Note the different horizontal scales.

With no mispricing there is no edge. Both expectations are zero to Monte-Carlo precision, which is the claim of §5 confirmed. The naked seller still wins 59% of the time, at zero expectation: negative skew in its purest form. The calendar wins only 43% of the time at the same zero expectation, because its losses are capped and its wins are not small: *the calendar turns a negative-skew payoff into a bounded one.*

With mispricing, the calendar retains about 11% of the naked edge. The calendar earns $\$0.38$, versus $\$3.33$ for the naked straddle. These are the gains obtained when the event variance is repriced from its implied level to the realized one, here a reduction to 40% of the implied variance. Repricing both trades at the lower variance reproduces these P&Ls to the cent. The 8% figure in §9.3 is a local exposure: it describes the response to a small change in event variance. Over a larger repricing, the curvature of the option values matters, and the calendar retains a slightly larger fraction of the naked edge, 11% in this example.

This is why a calendar should not be viewed simply as a cheaper way to capture the naked straddle's edge. The event is common to both legs, so most of the corresponding P&L cancels. What remains is the difference in how the two legs respond to the same event variance.

Per unit of tail risk, the calendar is better, before costs. Divide each trade's expected P&L by the size of its 1% loss: the calendar earns 0.41 per unit of that loss, the naked straddle 0.31. The calendar can therefore be traded in larger size for the same tail. About twelve calendars carry the same 1% loss as one naked straddle, and that book expects about 4.4 against the naked 3.3; its worst case is capped at about 10.8, while the naked straddle's worst simulated outcome is 32 and has no limit in principle. The advantage is thin: twelve calendars trade far more premium than one straddle, and §11.7 shows that costs above about a quarter of a percent per side erase it.

11.7 What execution costs do to it

The calendar's edge is small, and a small edge is exactly what transaction costs consume. Charging a cost c per side, as a fraction of the value of each option traded at entry and at exit, in the overpriced case the new expectations are:

Cost per side	Naked	One calendar	Calendars at equal 1% loss
0%	+3.33	+0.38	+4.35
0.25%	+3.29	+0.28	+3.27
0.5%	+3.24	+0.19	+2.18
1%	+3.15	+0.00	+0.02
2%	+2.98	−0.37	−4.31

The last column holds about twelve calendars per naked straddle, the number that gives both books the same 1% loss.

The naked straddle breaks even only at a cost of 19% per side; the calendar at 1.0%. The asymmetry is structural: the calendar keeps a tenth of the edge but trades more than twice the premium, four legs in and out instead of two. The tail-risk argument of the previous subsection is correspondingly fragile. Scaled to the naked straddle's 1% loss, the calendar book beats it only while costs stay below about a quarter of a percent per side, and crossing the spread on short-dated single-name straddles around an earnings release often costs more than that. The conclusion is not that the calendar fails, but that it belongs to liquid names and patient execution, working orders inside the spread rather than crossing it.

11.8 The long-leg tenor is the dial

The tenor of the long leg decides how much of the event it absorbs. Keeping the short leg fixed and varying the long leg, with its implied volatility consistent with the same σ_{diff} and σ_{event} , in the overpriced case:

Long leg	Debit (floor)	$E[\Pi]$	1% outcome	$E[\Pi]/ 1\% $	Exposure kept
5 days	0.38	+0.17	−0.38	0.44	4%
8 days	0.93	+0.38	−0.92	0.41	8%
13 days	1.78	+0.64	−1.74	0.37	15%
23 days	3.33	+1.01	−2.98	0.34	24%
43 days	6.00	+1.45	−4.49	0.32	37%
Naked straddle		+3.33	−10.69	0.31	100%

A longer long leg carries less of the event, so the calendar keeps more of the event exposure and more of the edge, at a larger debit and a lower edge per unit of tail. As the long leg lengthens the trade slides toward the naked straddle; as it shortens, toward a fully hedged position with almost nothing left to earn. A rule of thumb such as a long leg two to three times the short one picks a point on this curve; it is not a law.

12. Practical considerations

12.1 Strike

Strike both legs at the money and at the *same* strike: Result 10.1 depends on it. ATM straddles concentrate gamma and vega where the event mispricing lives, and avoid mixing an event view with a view on the smile.

12.2 Expiries

The short leg should expire as soon as possible after the event, where the event is the largest share of its variance. The long leg is the dial of §11: longer buys edge at the cost of a larger debit and more exposure to drift in the surface over the intervening weeks.

12.3 Greeks

- **Delta** is close to zero at entry for ATM legs, and drifts after the jump: the short leg, richer in gamma, moves faster. Rebalance if the position is held beyond the first hours after the release.
- **Gamma** is net short but bounded, because the long leg offsets most of it.
- **Vega** is where intuition misleads. A one-to-one calendar is *long* parallel vega, because vega grows with $\sqrt{T}$ and the long leg has more of it. But it is *short event variance*. The distinction matters before the release: a uniform rise in volatility helps the position, while a rise in the event component, which lifts the short leg more, hurts it.

12.4 Exit

Close shortly after the release, within the first post-event session. Holding longer adds diffusion and surface-drift risk that the trade was not built to carry, and exposes it to follow-on information from the event itself.

12.5 When the trade does not work

1. **The event is fairly priced.** Expected P&L is zero before costs and negative after them.
2. **Execution is not cheap enough.** The calendar breaks even at a cost of about 1% per side (§11.7). Where crossing the spread costs more, the edge is gone before the event happens.
3. **Event volatility reprices upward before the release.** The short leg's implied volatility rises more than the long leg's and the position shows a mark-to-market loss before the event occurs. The floor of Result 10.1 concerns the event outcome, not the marks along the way; if the position is forced closed early, the loss is realized.
4. **The post-event volatilities do not collapse together.** The floor assumes both legs trade at the same post-event volatility. If the long leg's implied volatility falls further than the short leg's, or the short leg's stays elevated, the floor can be breached.
5. **The diffusion term structure is not flat.** The extraction of §3.2 then absorbs its slope into the event estimate, and the trade is sized on a mismeasured edge.
6. **American exercise and dividends.** The bound is derived for European options at zero rates. On American single-name options, early exercise of the short leg after a large move and discrete dividends perturb it at the margin, and should be managed.

13. Synthesis

One identity, $\sigma_{IV}^2 T = \sigma_{diff}^2 T + \sigma_{event}^2$, organizes the subject. It prices the event: the term-structure cliff reads the event variance as the height of a jump in total variance, and two expiries that both contain the event still identify it. It explains the path: implied volatility rises into the event because a fixed variance is spread over fewer days, and collapses after it because the event term disappears. It explains why the path cannot be traded: under fair pricing every hedged position through the event has zero expected P&L, however often it

wins. And it says what can be traded: the level of implied event variance against the realized squared moves of the name.

The same identity then shapes the trade. A naked short straddle expresses the view at full exposure, with an unbounded tail. A same-strike calendar, with both legs alive through the event, is short only the difference in how the two legs price the same event. That difference is small, so the calendar keeps about a tenth of the edge; and it is bounded, so, under the assumptions of Result 10.1, its loss through the event cannot exceed the debit paid. Because what remains is small, the calendar also depends on execution: it belongs to liquid names, where the cost of crossing the spread does not consume the edge it was built to keep.

Place in the series. The decomposition is the additivity of variance behind the variance-swap notes, and the dependence of the jump's P&L on where spot lands is the path-dependence studied there. Real events are often skewed, the jump term of the skew-premium notes: skew changes how the variance splits between the wings, not the arithmetic. And the volatility-change-rate notes will show the other side of the crush: at an event the move in implied volatility is orthogonal to spot, so no delta hedge reaches it, and event days must be excluded when that sensitivity is estimated.

The unifying observation. The naked seller is exposed to the realized variance of the event at full size; the calendar is exposed only to the difference between how two options price it. The first is unbounded. The second is floored at the debit, and pays for that floor with most of the edge, which only cheap execution preserves. Between the two lies a continuum, and the tenor of the long leg is the dial that chooses a point on it.

References

- Dubinsky, A., Johannes, M., Kaeck, A., and Seeger, N. J. *Option Pricing of Earnings Announcement Risks*. Review of Financial Studies, 32(2), 2019. Extraction of anticipated earnings jumps from the term structure of implied volatility.
- Demeterfi, K., Derman, E., Kamal, M., and Zou, J. *More Than You Ever Wanted to Know About Volatility Swaps*. Goldman Sachs Quantitative Strategies Research Notes, 1999. Variance, its additivity and its replication.
- Brenner, M. and Subrahmanyam, M. G. *A Simple Formula to Compute the Implied Standard Deviation*. Financial Analysts Journal, 44(5), 1988. The ATM approximation used for mental arithmetic.
- Andersen, L. and Andreasen, J. *Jump-Diffusion Processes: Volatility Smile Fitting and Numerical Methods for Option Pricing*. Review of Derivatives Research, 2000. The jump-diffusion framework behind the decomposition.
- Rebonato, R. *Volatility and Correlation: The Perfect Hedger and the Fox*. 2nd edition. Wiley, 2004. Volatility-surface structure and models with jumps.
- Bergomi, L. *Stochastic Volatility Modeling*. Chapman & Hall/CRC, 2016. Dynamics of the volatility surface.
- Sinclair, E. *Volatility Trading*. 2nd edition. Wiley, 2013. A practitioner's treatment of earnings and event volatility.

- Taleb, N. N. *Dynamic Hedging: Managing Vanilla and Exotic Options*. Wiley, 1997. Jumps, discontinuities and hedging through them.
- Alexiou, L., Goyal, A., Kostakis, A., and Rompolis, L. *Pricing Event Risk: Evidence from Concave Implied Volatility Curves*. *Review of Finance*, 29(4), 2025. Concave curves before earnings as a signal of event risk and of its price.
- Glasserman, P. and Pirjol, D. *W-Shaped Implied Volatility Curves and the Gaussian Mixture Model*. *Quantitative Finance*, 23(4), 2023. Bimodal densities and the smile shapes they produce.